DASE7506 Mini-project 2 | Student ID: 3036837930

1. Task, question and reported method

Selecting Reliable LLM Reasoning with a Compact Numeric-Relationship Ensemble

Research question. Can relations between quantities in the question and the final candidate answer improve a compact embedding-based selector without relying on a deeper text model? The hypothesis is that numeric structure contributes information that fixed whole-solution embeddings do not capture. A second hypothesis is that lexical conditioning helps distinguish which numerical relation is relevant to the requested quantity.

The course bank contains 4,755 training, 798 validation and 467 test questions, each with four candidate solutions and fixed 1,024-dimensional embeddings. Any candidate with the correct parsed final numeric answer earns a point. Multiple candidates can be correct. This is answer selection, not a verification of every intermediate reasoning step. All fitted parameters use supplied training inputs and references. No external dataset, pretrained weights, generated answer, or online inference call is used.

The reported checkpoint combines a regularized linear selector, three small candidate-set networks, and four sparse numeric residual models. It obtains 752/798 (94.24%) on validation and 387/467 (82.87%) on the official test bank. It occupies 2,077,521 bytes and runs complete fresh-process prediction plus scoring in 5.35 seconds on the local CPU.

Component	Role
Linear selector	Normalized embeddings, answer agreement and parseability
Three set networks	Candidate, question-set and same-answer context
Numeric residuals	Question quantities versus candidate final numeric answer
Final decision	Masked argmax over the four original candidates

Evaluation scope. The test bank contains only questions with at least two different valid parsed answers; validation includes 443 easier single-answer groups. Consequently, the overall validation/test gap is partly a composition difference. On the 355 validation disagreement groups, the reported model is correct on 309 (87.04%).

2. Model and training recipe

For candidate j, the linear feature vector contains its L2-normalized embedding, vote count divided by four, and parseability flag. A question-centered softmax model is trained against the uniform distribution over correct candidates, with L2 penalty 0.005. Its inference score subtracts 1.5 times a conservative arithmetic-mismatch indicator extracted from the candidate text.

Each candidate-set network uses the same 1,026 inputs, standardized using training-only mean and scale (scale floor 0.01). A 64-unit first layer produces h. The second layer receives h, the question-level mean h, the same-answer group mean, their elementwise interaction h*group, and the two agreement/parseability inputs. A 64-unit layer and scalar output complete the network. Dropout is 0.1; optimization uses AdamW (learning rate 0.0003, weight decay 0.01), batch size 256, gradient clipping at 5, and normalized correct-versus-incorrect pairwise logistic loss. Seeds 17/29/43 use 15/10/10 epochs. Network probabilities are averaged.

The base probability is p_base = 0.7*p_linear + 0.3*mean(p_network). Numeric residual features compare a candidate answer to up to eight distinct question numbers.

Residual view	Feature scope	L2	Weight
full	Flags and question unigram/bigram interactions	3.0	0.25
no_text	Numeric flags without word interactions	0.3	0.375
no_formula	Non-reachability flags with question words	0.1	0.125
target	Flags with requested-clause words	1.0	0.125

Each candidate vector is L2-normalized. The final score is log(max(p_base, 1e-9)) plus the four weighted residual scores. Unparseable answers are masked whenever a valid answer exists; ties follow the first maximum in bank order.

Residuals are fitted on training-only out-of-fold (OOF) base probabilities. Three partition seeds, each with three folds stratified by answer agreement, provide three probability estimates per training question. Each base model sees only the other folds. Their averaged matched-configuration probability is the fixed offset in a pairwise logistic objective with L2 regularization. Full-training base models are then used for inference.

3. Validation search and controlled evidence

Validation was the development comparison set. A first fusion search trained 36 residual models and evaluated 14,003 configurations by seeded random search and coordinate refinement. A subsequent joint search covered seven linear penalties, four neural fractions (0.1, 0.25, 0.3, 0.5), and three arithmetic penalties (0.5, 0.75, 1.5). Every one of its 84 base configurations used matched training OOF probabilities to retrain the residuals. It performed 70 linear fits and 3,024 residual fits, evaluating 79,012 configurations.

Linear L2	Best validation correct / 798
0.0001	749
0.001	749
0.003	751
0.0056234132519	752
0.01	751
0.1	750
1.0	749

Changing the linear regularization and matching the residual-training offsets did not exceed the previous 752-correct validation result. The best region retained the original linear penalty, neural fraction 0.3 and arithmetic penalty 1.5. At the leading matched configurations, replacing stale-offset residual weights with matched-offset weights changed no validation choices.

A fixed-weight leave-one-component-out ablation was performed on validation for the reported checkpoint. The full model improves over its matched base by seven validation questions.

Fixed-weight ablation	Correct / 798	Hard correct / 355
base_only	745	302
full_model	752	309
without_full	752	309
without_no_text	746	303
without_no_formula	751	308
without_target	752	309

The no-text numeric component has the largest observed marginal contribution: deleting it loses six correct questions. Deleting the non-formula component loses one. Removing either the full lexical interaction view or the target-clause view changes no correct-count total.

4. Results, resources and reproducibility

Finally configurations were evaluated with the official scorer.

Validation correct / 798	Test correct / 467	Test accuracy
752	387	82.8694%

Local hardware: Intel Xeon Gold 6248 and one NVIDIA RTX 3080 Ti (12 GiB). All inference is CPU-only with four threads across numerical libraries; the timing includes separate fresh prediction and scoring processes, imports, input loading, feature extraction and every ensemble member. Measured checkpoint evaluation took 5.35 seconds; independently retrained checkpoint evaluation took 5.40 seconds.

Resource / verification	Measurement
Inference assets (uncompressed)	2,077,521 bytes; approximately 1.98 MiB
Final-only training, 44 fits	98.71 seconds
First fusion search	63.00 seconds; 14,003 configurations
Joint search and validation	1047.85 seconds; 79,012 configurations
Fresh CPU environment	Python 3.12.3; NumPy 2.4.6; SciPy 1.16.3
Original/reproduction training stack	PyTorch 2.12.1+cu130; CUDA GPU

AI assistance and reuse

OpenAI Codex provided assistance with method design, implementation, experiment execution, hyperparameter searches, diagnostics, packaging and drafting this report. No external pretrained weights or additional data are used.